

ActInflab Livestream #035.0 ~ "A tale of two architectures free energy, its mod

Livestream | 1:25:52

Welcome Blue, I'm about to play the theme song.

Hello everyone, welcome to ActInflab livestream number 35.0.

It is January 4th, 2022, so also, Happy New Year to everyone.

Welcome to the ActInflab.

We are a participatory online lab that is communicating, learning, and practicing applied active inference.

You can find us at the links here on this slide.

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Today in ActInflab livestream number 35.0, we are going to be learning and discussing this paper,

A Tale of Two Architectures, Free Energy, Its Models and Modularity by Majid Beni in December 2021, but kind of published in 2022.

Today in ActInflab is a series of videos that we have a series of videos that we have in the past.

We are going to be looking forward to the two upcoming weeks when Majid may join us, and hopefully many of you will join us, and we are just going to have a good time, talk about the paper, what have we learned from it, and what are the implications.

Today we are going to be more focused on the aims and claims, the abstract, the roadmap, the background, the keywords,

and there aren't any figures or formalisms, but we will go through some of the key arguments, which are very provocative and very salient.

So, if you want to get involved, just get in touch, and we will let you know how to participate.

Let's begin with a little bit of an introduction and a warm-up.

So, feel free to say hello and maybe just one thing that made you excited to discuss this paper.

So, I'm Daniel. I'm a researcher in California, and I was excited to read philosophical work by Beni, who always writes very interesting things, and just understand how architectures of physical systems and their functional descriptions are related to the FEP. Dean?

Thanks, Daniel. I'm Dean. I'm here in Calgary. It's really, really cold and snowy, but that's not why we're here.

My real question was, so what do we do about this seeming philia or attraction to the ability to categorize?

And so, the question is, do we model up? And yeah, so reading the paper, I kind of tried to figure out the answer to that.

I'll pass it down to Blue.

So, I like this paper from the perspective of multi-scale dimensionality and also modularity, like modules at different dimensions,

the scale-free aspect of the Markov blanket, or the not scale-free aspect of the Markov blanket.

And anyway, I thought it was just really neat and a cool way to fit the pieces together.

Excellent. So, we have some slide with things that we want to discuss in the dot one and the dot two and beyond.

So, at any point, feel free to add information there and we'll be writing things there.

So, we're not going to get to it all. We're not going to make this an 11-hour video.

We'll get things down to discuss in 35.1 and two.

But we can lead in with the big question, a big question, which is, how do we think about modularity of different kinds,

like functional, effective, and anatomical, which we'll be discussing later, in active inference and free energy principle?

And why does that matter? Why does it matter how we think about modularity of different kinds? What does it mean?

Here on this slide are some images related to just different uses of modularity.

Just modules like Lego bricks.

Other times, modularity can refer to like part of a network that's more densely connected, like a click.

Other times, modularity doesn't refer to hot-swappability, but rather to a larger structure that's composed of smaller modules,

whether they're the same or different, it's often applied to mathematical systems or biological systems.

So, there's so many aspects of modularity.

And today, we're going to be exploring, what are those aspects of modularity and why does it matter for the free energy principle?

Blue or Dean, like any thoughts on the general modularity question?

I got a quick question. So, for me, when we see these sort of images that we externalize and artifacts and that sort of thing,

when I started getting into the paper, the real problem for the modularity argument is,

what does localize mean when we're talking about fMRI representations as the tool?

Does a bump in electrochemical impulse mean a clearly defined border?

I don't think so. Of course not.

Where one bump ends and another begins all inside the same energy field might be captured through fMRI imaging.

And is that misleading because there's no moment of stasis because the magnets inside our heads keep boring constantly?

So, that was, again, another question that I sort of tried to carry on through as I was reading what the author wrote out.

Yeah. Where are we talking about modules? Where are we modularizing?

And does that map onto potentially realism and instrumentalism?

Any big setting ideas, Blue, or we can continue to aims?

All right. So, the paper that we're discussing is by Majid Beni, The Tale of Two Architectures, Free Energy, Its Models and Modularity.

And it's in the paper, Consciousness and Cognition.

Just to go through some of the key aims and claims, and then feel free to add a comment.

So, the paper presents a model-based defense of the partial functional informational segregation of cognition in the context of predictive architecture.

More specifically, to defend the modularity thesis, the paper confutes two counterarguments that lie at the center of Hippolyto and Kirchhoff's 2019 recent confutation of the modularity thesis.

To confute is like to refute.

The main insight of the paper is that Hippolyto and Kirchhoff's counterarguments miss the mark because they dismiss a few rudimentary facts about the model-based nature of dynamical causal models and Markov blankets.

So, that's why we kind of let in with the big question about modularity.

Why does it matter?

Why does it matter?

Because we're kind of down the road a little bit.

Like, okay, modularity matters.

It matters how we think about and analyze and design modular systems.

And then, how does it apply to FEP?

And Hippolyto and Kirchhoff make a confutation of the modularity thesis, which we can look at soon.

And then, Beni is firing back with this current paper.

So, that situates us like in the biggest scope of thinking about modularity, H&K 2019, and here we are with MDB 2021.

So, can I add to that, Daniel?

Mm-hmm.

Yeah, so, we didn't put encapsulation in here, but I know we're going to talk about that a bit.

But in order to be able to make sure that we don't mix things up, we have to understand that encapsulation is a separation of what matters or what we consider to be a signal from the noise.

And I always think back to Nate Silver when I try to compare those two things.

So, it's a momentary on-off or it's a momentary in-out.

Not that there's a walled-off section, per se, in the brain like we can diagram out or represent as an artifact.

FMRI images, however, might coerce us into thinking that that's what's happening, that there's actually a walled-off section, when that isn't actually the case.

To avoid the cognitive overload, there are brake stop mechanisms in our mind.

We don't have to tell our pancreas or our basal ganglia what to do, which then assists us in our ability to take the external states like these slides in.

And I think that's what Benny was trying to point to, although he didn't probably go into that explanation the way that I just did.

But I'm kind of in defense of this author, so I'm starting off right now.

I'm in his corner. I'm ready to tag up and climb in the ring.

So, just to kind of piggyback onto that, like when you're talking about encapsulating, it's really talking about drawing a boundary around something.

And like, isn't the Markov blanket, like isn't that the ability to draw a boundary around something, right?

So, I don't know, like this encapsulating, Markov blanketing, boundary drawing, like all of those are conflated in my mind.

And I don't know if there's like a separate way to tease them all apart or if I have correctly grouped them.

Great. Onwards. So, I'll read the first two and then blue for the second two.

The paper presents a model-based defense of the partial functional informational segregation of cognition in the context of the predictive architecture.

The paper argues that the model relativity of modularity does not need to undermine its tenability.

In fact, it holds that using models is indispensable to scientific practice.

And it builds its argument about the indispensability of modularity to predictive architecture on the indispensability of scientific models.

More specifically, to defend the modularity thesis, the paper confutes two counterarguments that lie at the center of Hippolito and Kirchhoff's 2019 recent confutation of the modularity thesis.

The main insight of the paper is that the main insight of the paper is that Hippolito and Kirchhoff's counterarguments missed the mark because they dismiss a few rudimentary facts about the model-based nature of dynamical causal models and Markov blankets.

Great. So, nice philosophy abstract.

Here is the roadmap where we are going. And so, this has a fun roadmap. There are nine sections.

There's the introduction. And then there are eight sections with, or dare I say eight modules, that have kind of interesting names that we'll be looking into.

And there are not any figures or formalisms, but the sections are here. Maybe Dean will rejoin.

And welcome back, Dean.

C'est la vie. I need an Ethernet cable from my heart to yours.

So, the keywords are free energy principle, model-based science, modularity, and then there's DAG, directed acyclic graphs, and then DCM, which is going to have two different unpackings as directed cyclic models and as dynamical causal models.

And we're going to go through these keywords kind of as they appear in the paper. We're not going to have a separate section.

So, let's go right into the introduction section of the paper where we find somewhat of an epistemic preamble.

So, not to read the whole thing, but, Benny belongs to the clan of philosophers with Carnap and Kuhn, who concede that the choice of theoretical frameworks is not completely arbitrary, because some pragmatics affect the choice of theories.

Okay, we have dispensed with the naive postmodernist perspective.

And more generally, Benny identifies as a naturalized metaphysician in the sense of Ladyman and Ross, in that metaphysics is worthwhile when it is informed by our best scientific theories of the field.

So, it's kind of just like a classy, co-informing process between science and metaphysics, and it is drawing on

this work by Ladyman and Ross,

Every Thing Must Go, Metaphysics Naturalized.

And then just to look at the chapter titles, because it's not a book that I was very familiar with, they're defending science.

They're defending a scientific realism and a structuralism.

So, maybe we're not going to be in touch with the ultimate reality, but at least we can be in touch with something that's constructively empirical and scientifically real.

They talk about the unity of science, consilience, like E.O. Wilson, and about causation in a world that's structural, but also complex.

And so, that's sort of the preamble where Benny characterizes his own stance and kind of gives this background that is quite personal in some ways, but then will lead into our understanding about the paper and the contributions.

So, anything to add on that?

All I would try to quantify here is that he's not starting from a position of, did you see that?

He's basically starting from a position of, did you see what I see?

He holds that as being valuable.

Great.

That takes us to model-based science.

So, as we saw in the chapters of Lady Men and Ross, scientists use models.

It's kind of like saying we use maps on territories.

And there's so many kinds of models.

A few that we've seen are, for example, a graphical visualization of a Bayesian statistical model.

And one example of that is like this partially observable Markov decision process that we often look at.

And what do models do?

Well, that's quite a big topic, but they represent hypotheses as well as useful models of conceptual or empirical systems.

And so, there's many, many kinds of models, and some of them can be reflected graphically.

And in that sentence, graphical is referring to visual, like a computer graphic, but also graphical, like a graphical network.

And so, we're going to be talking about the visual aspects of graphs, but this isn't a study in aesthetics.

This is going to be talking about the topology of graphs in their network sense.

But model-based science is just the kind of pragmatic perspective that scientists use models, implicitly and explicitly.

And there are certain things that we can say about those models.

Anything that either of you would add on model-based science?

I'd want to know who disagrees?

Who doesn't think that scientists are using models?

Like, is this just a total meaningless claim?

Or are there controversies related to this?

Okay.

So, let's look at a few different types of models, specifically these graphical models that can be represented by networks.

So, one kind of special graphical model is called a directed acyclic graph, or a DAG.

And so, DAGs, as their name would have it, do not have cycles amongst variables.

So, nodes are variables, and edges are directed influences, and there are no cycles in DAGs.

And so, it turns out that computing DAGs, it's relatively efficient, and it has known algorithms.

So, here is an image of a DAG that I got from a 2012 paper in Ecology.

And so, the parameters are defined here, and you can look for more details.

But just visually, these are different modules of the model.

There's, like, this underlying process model.

There's the individual covariate model.

There's an observation model.

Whatever those things are.

But one can imagine that you can, like, start at the parts that don't have any parents, and then kind of drip coffee your way to N and Y, which are, like, the ultimate grandchildren nodes.

And then N and Y don't feed back to change the other upstream parameters.

So, DAGs are not all of this exact shape or type, but what makes them in a category is that they don't have cycles in the way that the variables are influencing each other.

That makes them easy to compute and relatively straightforward.

We can contrast that with directed cyclic models.

So, we're still talking about graphical models here.

Directed cyclic models, or DCM, have causal loops in the graphical models.

And so, this can make certain kinds of computation more difficult.

I'm sure people can think about different ways that that might happen.

The direction of the edges determines the influence of one random variable on another.

So, if it doesn't have cycles, it's a DAG.

And inference on those DAGs may be performed exactly using algorithms such as belief propagation or variable elimination.

Like, you can kind of collapse a subtree into a simpler system.

But with cycles, it's a little bit more complex.

So, that's the difference between the DAG and the DCM, directed cyclic model.

But then there's going to be another DCM coming up later.

Any comments there?

Do you think that the looping can begin kind of a spiraling, kind of an accumulation effect that the DAG model,

which with its narrowing and narrowing and sort of elimination focus...

Well, Benny talks about the fact that they're actually complementary.

But do you actually think they're complementary as well?

I think that they exhaustively describe the landscape of graphical models because either it has a cycle or it doesn't.

But it's quite literally the difference between, like, dropping a coin in a little machine that just dribbles down and then playing pinball where you could have all kinds of feedback.

So, I think they're mutually exhaustive.

But that may be sort of a deflationary stance because they're defined as contrary to each other.

So, we're not going to go too into detail here.

But sometimes you want to use a model that is acyclic even for a cyclic system.

For example, within one time step, it might be a heuristic approximation to make it a DAG model, directed acyclic model, rather than a cyclic model.

So, we're talking about our statistical models here, not the topology of how things are connected in the real world.

But I just thought it was very interesting to talk about Bayesian networks.

And often, Bayesian networks are framed as only DAGs.

So, the standard definition, according to these authors, the standard definition of Bayesian networks is usually one-directional acyclic directed graphs.

In higher dimension systems, further probabilistic relations can be recovered from network cyclic properties and the joint probability of all variables.

And this paper shows, like, for a two system, where there's just a kind of back and forth between these two x and y variables.

Here's x , y , and z in a three-way loop.

And then here's a four-way loop.

And then they also look at n -order loops.

So, it can generalize.

So, what was once only imagined is now proved.

In other words, let's not get too confident about, like, what can or can't be done with one type of model family or the other,

because it's just one clever paper away for there to be an approximation or to be an exact solution.

Now, we turn to the other sense of DCM.

And this is dynamical causal models.

And so, these DCM can have graphical structure that also is directed acyclic graph.

But it's a different usage of the term.

And so, DCM, here we could point to many resources, like Friston et al., the 2007 Statistical Parametric Mapping Textbook,

or a later paper of Friston et al. on network discovery with DCM.

And so, I thought that 2011 paper was a little bit more relevant,

because it's about specifically inferring or discovering the functional architecture of distributed systems using dynamic causal modeling.

What does this technique do?

It uses Bayesian model selection, so comparison of the relative adequacy and complexity of different Bayesian models,

to identify the sparsity structure, which is where there are edges or not,

or where there aren't, which is the same thing as where they are,

in a graph that is best explaining in observed time series.

The implicit adjacency matrix, that's that sparsity structure,

specifies the form of the network, e.g., cyclic or acyclic.

And then later on, we see, unlike conventional approaches to network discovery,

DCM permits the analysis of directed and cyclic graphs.

So, DCM is an approach for looking at a time series, that's the dynamic part, through time,

and then looking at how variables statistically cause each other.

And so, a statistical cause is like when one thing happens,

and then at some later time point, there's another thing happening.

That's a statistical causal model, like Granger causality.

We're talking about our model.

So, if Bitcoin price goes up, and then later oil price goes up across the whole time series,

so there would be a directed causal edge,

there might be a bidirectional causal edge or a one-directional causal edge,

it doesn't mean that in the real world, one going up caused the other.

It's just talking about our time series data.

So, that's dynamic causal modeling, and it can be applied to neuroimaging data.

Blue?

I have a quick question here.

So, I get the time series, if that makes it dynamic,

but I also thought you could have functional studies

that also fall into the category of dynamic causal modeling,

like gene knockouts, right?

So, you knock out a gene, not necessarily a time variation,

but a variation in the graph itself,

and so you knock out a gene, and then isn't that also dynamical causal modeling,

or that's something different?

One could have the nodes be gene expression through time,

and then test how perturbations or counterfactual network architectures

would result in different expressions through time?

But can you take the time out of it and have it still be a dynamic model?

I guess it's just snapshots, right?

So, it's not time series, but like gene series.

I don't know, like knocking out the genes in the series to make it a dynamic model.

If it were just a single time slice,
then it's a little bit unclear what the edges would reflect,
because cause is like effect through time in the statistical sense.
So, if it were just like, I have like, you know,
there's 50 people here and 20 people here and 10 people here,
it's hard to say where's there a causal edge.

Whereas if all three were changing, then one could draw these causal edges.

Okay.

Now we turn under the scope of the big question about modularity to specifically modularity and the free energy principle, FEP.

Okay.

Okay.

So, this is the big topic and kind of the entree that Benny is bringing to us to discuss.

And it's highlighting a really important issue,

which is the conceptual compatibility of modularity with the free energy principle.

And it's a speculation that this conceptual point may well become significant in the course of scientific development of the FEP.

So, this is like a philosopher engaging in a scientific discussion,

because there actually is a lot on the table if we get modularity right or wrong.

So, what is the relationship between FEP and modularity?

Is it compatible?

Is it necessary?

Sufficient?

Incompatible?

Like, if we have modularity, do we get FEP?

Right?

That's sufficient.

Do we need modularity for the FEP?

Necessity?

If we have it, can we not get the FEP incompatibility?

Are we talking about the modularity of our world?

Carving up nature at the joints?

Or of our scientific statistical models?

Instrumentalism?

Carving up our models at the joints that we designed?

And then how does our stance or our perspective on this topic of modularity and the FEP influence our generative model?

How we think and act?

Whether we're the ones collaborating and writing scientific papers or not?

And then just to add one more note, despite the great unifying scope, FEP has been criticized on account of not discussing the functional mechanisms of the brain.

So, the unifying scope is kind of like top-down.

That's the Rampstead et al., 2018, all of space and time under the FEP envelope.

That's the top-down description.

And then the functional mechanisms are like those minute particulars.

That's the bottom-up components that build the system.

And that sort of meets in the middle with the real-world systems that people are always discussing.

So, we're talking about how modularity works with the FEP or not.

Any thoughts, Bluerdeen?

Well, only thing I would add is that you can be strictly top-down or you can actually, as Benny would point out, keep an open mind, which means that you've got some movement for both top-down and bottom-up.

So, we'll probably get into that more in the point one than point two.

Yes, exactly.

Exactly.

We're now going to go to some classic citations.

So, Benny writes,

To address this problem of modularity in FEP, FEP theorists may want to provide finer details about the functional mechanisms that enable FEP.

So, that's how we're going to make a comeback to Colombo et al., 2021, Colombo and Wright 2016, is by providing finer detail.

In addition to reinvigorating its unificatory scope, unifying scope, maybe, that would be in line with Fristen's preceding remarks that, despite its roughness, FEP may be developed to provide a guide to brain functions.

So, we're going to do both.

We're going to keep the big scope and we're going to provide details.

So, let's look at some of those details.

This is also going to provide a really good introduction to the difference between the functional and effective connectivity.

And connectivity here is referring to statistical measurements coming from brain regions.

Because anatomical connectivity is just, are the pieces physically connected?

You know, the knee bones connected to the leg bone, that kind of descriptive anatomical connectedness.

First in 1994, functional and effective connectivity in neuroimaging, a synthesis.

So, back when we were younger, weren't we?

The paper reviews the basic distinction between functional and effective connectivity in neuroimaging.

Going over to the origins and definitions.

Functional connectivity is defined as the temporal correlations between spatially remote neurophysiological events.

So, that's not the only way to say it in English or any other language.

But we're not talking about a philosophical sense of function.

Like, what does it mean or what is its teleology?

We're talking about statistical variation in measurements.

This definition is operational.

Very cool.

And provides a simple characterization of functional interactions.

The alternative is to explicitly refer to effective connectivity.

I.e. the influence that one neuronal system exerts over another.

So, temporal correlation is things that are, like, they could be uncorrelated in the micro,
but then they're both going up together.

That would have functional connectivity.

Whereas, effective connectivity would be looking at how changes in one at the next time step influence changes in the other.

And so, those could be, like, going in different directions at the macro,
but they could have effective connectivity because they influence each other statistically.

Let's stay with this Friston 94 theme, but we're not going to go through the whole thing.

This is just to show, like, what was being discussed and explored and characterized about 30 years ago.

So, keep that in mind.

We have a long pheromone trail to catch up with Carl on.

They had models way back in 1994?

This is 94.

Yeah, they had models back then?

Well, they were made of wood, but yes.

This is a statistical parametric map, hashtag SPM, projected onto a brain.

And on this map of the images taken through time of a brain that were fMRI in this case,
one can take that data object and find the eigenvalues.

And there are mathematical relationships between functional and effective connectivity.

Not going to go into it, but it's an awesome topic explored here and in several earlier papers and in the 2007 SPM textbook and in later papers.

What was able to be said even way back then?

Well, this is one of the figures.

These are looking at the loadings on different vectors.

So, it's a spatial mode that reflects an eigenvector.

So, it's kind of like a vector through space and time activity that explains a large proportion of variance in the data.

And the first vector, that's this top one, is clearly related to the difference between word generation, even numbered conditions, and word shadowing, odd number scans.

So, this was an experimental paradigm where the person was being asked to do two different things alternating.

Generate a word and then shadow a word.

Like, I guess, repeat a word that was already said.

And then, this vector, which had this activation pattern on the right in the brain, was negatively correlated with the odds and positively correlated with the even.

So, it's a plausible mechanistic statement about the differences in the activated neuroanatomy.

It doesn't mean it's simple to interpret, but it's a vector whose expression is correlated with something in the experimental design.

Like, if you did a double-blind study and these people got the drug and these people got a different drug, the correlation between them would be ascribed to the difference that you as the experimenter imposed.

Whereas, here's an example of another vector that declines throughout the entire experiment.

And so, that might reflect, for example, attention dropping off through time.

And so, that's a lot of neuroimaging then and now looking into these summarized statistical representations of activity patterns through time.

And also, for later, just, it's cool to look at some of these citations, like the Edelman citations, Tonini and Sporn's 92.

These are really classic citations.

So, there we were in 94.

And then, this is a dream of the future from 94.

So, here's Zaragami and Friston in 2020.

Again, talking about effective connectivity, biologically grounded models like dynamic causal modeling and dynamical systems theory.

So, improving and iterating on those basic ideas.

And then, also, what does it look like to apply it?

And what kinds of conclusions and statements do scientists make when just applying these methods?

So, for example, this paper was in 2019.

And it is called Dynamic Causal Modeling of Effective Connectivity.

Should sound familiar.

During Anger Experience in Healthy Young Men.

Seven Tesla MRI Imaging Study.

And so, you can read the abstract.

But they find that viewing certain kinds of films leads to activation of certain brain regions.

Dot, dot, dot, dot, dot.

We propose a model of effective connectivity associated with the anger experience based upon dynamic causal modeling.

The findings have implications for psychiatric disorders.

So, that's what people do.

They put people into scanners.

They take measurements of the hemodynamics of the brain.

And then, make a statistical model of it.

And the edges that exist across all subjects or only in the ones who have this diagnosis or only in these experimental conditions.

Those edges are interpreted in a certain way.

That's what's on the table when we're discussing all these techniques.

Any comments on Fristen94?

Or we'll go to the other classic.

Just that effective connectivity is what I was thinking of, like, in the gene network.

Like, how are they effectively connected?

So, it's, like, makes sense now.

Yes.

Another classic, but a more recent one, is 2009.

Fristen's paper.

Again, single author paper.

The free energy principle.

A rough guide to the brain.

So, now that's...

How long ago?

13 years ago?

The free energy principle might provide a comprehensive account of how we represent the world and come to sample it adaptively.

Through action.

By moving our little eyes around.

The FEP provides a mathematical specification of what the brain is doing.

Now, that starts to sound like realism.

Right?

What the brain is doing.

Not a functional description of the models we make about the brain.

It is suppressing free energy if it uses gradient descent.

And so, the box with the questions for future research are really still key.

What optimization schemes does the brain use?

These are, like, still important questions.

And then also, the domains of application.

It's just always good to see, like, what was the FEP's description and scope 13 years ago.

Some of the images we've even seen again and again.

With sometimes some variance.

But these are, like, images that we see commonly.

We talk about action to minimize surprise.

And perception to optimize the bound.

So, this is, like, the predictive processing element.

And this is, like, the control theory element.

We talk about the topology of perception, cognition, and action.

So, suffice to say that there are equations, figures, and it's another great paper on the long road towards the FEP.

Any comments on 2009 paper?

All right.

Blue, would you like to describe modularity?

Sure.

So, I just wanted to explicitly define it.

Because we talked about it a little bit in the abstract.

And we're going to talk about it a lot more.

But in the paper, the author defined modularity as information encapsulation, which Dean mentioned earlier.

And he says, informational encapsulation holds that when engaged in information processing, the subsystems do not have access to information processing in other compartments.

And information processing within each subsystem is constrained within the computational mechanisms of that subsystem.

And so, the modules are then defined by information segregation.

And it's interesting to compare modularity, like, is encapsulating information in a module, versus defining individuality, which, both, like, Mike Levin in the computational boundary of a self paper that we did last summer,

and then David Krakauer in his 2020 theory about the information theory of individuality, both of those authors talk about information sharing, like, information sharing, like, with cells to form a tissue, for example,

in Mike Levin's work.

And then in the information theory of individuality paper, like, up and down across different levels, so that you can define individuality at any given level.

But it talked about how information is shared with this bidirectional causation in that paper.

And so, I think it's interesting to compare this idea of informational encapsulation in a module with these definitions of biological individuals.

Very cool.

Like, is modularity the same thing as individuality?

How about continue describing, maybe, in this discussion on intransitivity?

Well, so, I had to, like, look this up.

So, I just, like, I went into the paper, like, I just wanted somebody to just define, what does this intransitivity mean?

And so, the author says,

the general insight is based on the intransitivity of the passage of prediction errors

between levels whose information can affect one another,
only in cases of uncertainty and noise.

So, I feel like that this is, like, intransitivity of information between the modules.

Like, but it was just not clear enough for me.

So, then I tried to look up, like, what is intransitivity?

What is this intransitivity theory?

And I came across this really cool paper, which is 2015 and entropy,
intransitivity in theory and the real world.

And I think it's really different than, like, what we're talking about here.

But it talks about, like, intransitivity in terms of preferences.

And it said the choice between transitive, which are transitive, which are absolute,
and intransitive, which are relativistic models,

depends on the nature of the processes that these models are expected to reproduce.

Many people, however, have psychological difficulties in accepting a relativistic approach,
expecting an absolute scale of judgment from bad to good,

which can be suitable in some cases, but excessively simplistic in others.

And they gave the idea of intransitivity in game theory.

And so, like, an intransitive game is just rock, paper, scissors,
which obviously, like, your choice.

There's no, it's not rock is always good, paper's always good, tails never fails.

It's always relative to what the other people are choosing, right, in this game.

And then they really go into, like, multidimensionality and intransitivity.

So he says in the paper,

coarsening, which is like coarse graining,

in multidimensional cases become strictly intransitive.

And the cases without strict intransitivity are degenerate.

Either dimensions are redundant or coarsening is performed

after merging the fractional variables into the overall utility

instead of independent coarsening for all or some of the criteria.

From a philosophical perspective,

this statement can be presented as a continuum argument for intransitivity.

Small alterations are commonly overlooked for secondary parameters

that can be accumulated into critical differences.

And so I just thought that this was cool

because it talks about, like, philosophy

and some things that we're talking about

and relative or, like, you know, realist versus instrumentalist,

but it's completely unrelated.

And I just wonder, like, is it completely unrelated?

Or can we go back and then, like, link intransitivity

full circle into what we're discussing here today?

So it was just, I put this in here

because I thought it was a cool paper.

And in my search for intransitivity, that's what I ran into.

Awesome.

Putting a number on something often makes it seem like

whatever you're putting the number on, they are transitive.

You know, if everybody gets a score out of 100,

then there's a natural order.

Or if we're ranking policies according to their expected free energy,

then there might be some ordering of policies

according to which ones are best minimizing free energy.

And I think this idea of intransitivity within the graphical model,

what kinds of information are being passed,

and what does it mean?

And then just more broadly, like you brought up with the game theory,

it's very interesting.

All right.

Let us continue almost finishing the first section

and get to modularity, FEP,

and the target of the paper,

which is the Hippolito and Kirchhoff paper.

So, without intending to map the course of the evolution of the FEP,

like we will, Blue,

I will just remark that the assumption of modularity

has been fruitful in the study of brain functions.

And so, keeping an open mind about that assumption

could be fruitful for the FEP too.

Some significant critiques,

such as Hippolito and Kirchhoff 2019, notwithstanding.

If you want to read more,

definitely check out this paper from late 2019

by Ines and Michael.

And they are also, as well as any of you,

welcome to come on and discuss this paper.

That could be really cool to like kind of embody

and enact the paper.

And are we on the same page?

Or what are our perspectives,
even if we are on the same page?

And this paper is engaging in the discussion
of the relationship between modules and modularity.

The main goal of the paper is not to defy
Hippolito and Kirchhoff's attack completely.

The notion of modularity that Benny will defend
is weaker than the notion that they attack.

So, what senses or types of modularity are we dealing with?

What are the stronger and weaker claims?

And we'll look at their definition on the next slide.

The main point of the discussion
is that following Hippolito and Kirchhoff,
the modest definition of modularity
is neither insignificant nor indispensable.

So that made me ask,
are we seeing a trend towards instrumentalism,
bottom-up, weak claims,
like the entailment, exploration of 34,
and deflationary models,
and moves away from naive or realist, strong claims?

And so that just totally points to the role of philosophy
in an ongoing scientific domain
because things like strong and weak emergence
or entailment relationships
and these discussions on modularity,
they really matter for how science is done
and how it's interpreted.

So it's just really cool to be seeing that happen
in a new way.

Again, we're not going to go into the whole argument
of Hippolito and Kirchhoff.

Check out the paper.

But they are going to consider
the three most well-articulated,
which is funny because that means well-said,

but also well-jointed,
arguments for the view that modularity
and predictive processing work well together.
So predictive processing here is also referring
to kind of that component of active inference.
And they're going to argue
that all three of these arguments for modularity
come up short, albeit for different reasons.
So they kind of do like a steel person
where they give the strongest argument for that tack,
and then they refute it.
And the three arguments they discuss
are the epistemic Bayesian courtroom argument,
which is just the idea that courtrooms are modular
and cognition is kind of like a courtroom,
like the eye is like a witness,
and then there's like another witness thereof.
And so if that's modular,
then shouldn't predictive cognitive architectures
be modular?
B, the intransitivity argument,
which is that the causal influences
across hierarchical levels
are instantiating mechanisms
that implement causal Bayesian networks.
And they basically say
that is pitching predictive processing
in terms of the DAG
as opposed to cyclic models.
And C, the Markov blanket argument.
The notion that a Markov blanket grounds the idea
that predictive processing exhibits modularity.
And so they are going to explore those
and confute them as they say.
And one of the key points
is when cognitive neuroscience
works with an acyclic Markov decision scheme,
so that's like the partially observable

Markov decision process
that we were talking about earlier,
it may very well be methodologically misguided.
Why?

Because there's an increasing tendency
within neuroscience
to emphasize recurrent
and reciprocal neuronal processing
within the newly emerging
dynamical causal modeling frameworks.

So, how are we going to think about
the fact that the brain
does have these feedback loops
and square the circle
with the DAGs and the DCMs?

And that's kind of the introduction.

That's where we're set up,
which is that Hippolito and Kirchhoff
have framed and then attacked
three pro arguments
for modularity and cognitive architectures
related to predictive processing.

Benny is going to kind of
take some of that energy
and with a weaker version
or a little bit of a different version,
not simply stronger or weaker,
because that would actually imply
transitivity of strength,
but with a slightly different tack
is going to recall us to life
through a discussion of modularity.

So, this section
is only going to have one slide on it,
and then either of you,
please feel free to add any comments.
There's been a long history of debate
on modularity in neuroscience.

And Fodor,
who's a very influential
cognitive scientist,
defined modular cognitive systems
as systems that are domain-specific,
innately specified,
hardwired, autonomous,
and not assembled.

That's not exactly the definition
of modular
that we've explored
in some other slides,
but the words are what they are.

And that is the Benny quote
about that functional isolation
between modules
is what defines them.

So, it's kind of like
when there's more connection
within than between,
then that's modular.

And so, Benny writes,
the present paper
does not intend to defend
Fodor's version
of the modularity thesis fully,
probably because some of these pieces
about like innate specification,
hardwiring, autonomous,
unassembled.

Take it or leave it.

It was the 80s.

But Benny is simply aiming
to show that some critiques
of modularity
do not need to undermine
the possibility
of informational encapsulation.

And so,
although a falsification
of informational encapsulation
leads to the renunciation
of modularity,
a defense of partial encapsulation
would not establish
a full-blooded form
of modularity.

So,
if you refute encapsulation,
you have to renounce modularity.

However,
parcel
or transient encapsulation
doesn't give you
strong modularity claims.

However,
it can be compatible
with the instrumental usage
of models
that have modularity.

So,
those are some of the main arguments
that set Benny up
to defend a version
of modularity
that's quite modest.

And it's almost like
after that kind of a modest
deflationary take,
the question is really,
is there any value
in that definition
of modularity at all?

So,
the debate isn't over
whether we can discuss

things modularly.

If we're going to take

this different tack,

then really,

the question is like,

does it matter?

Any comments on section two?

Some.

I don't know.

I don't want to get too deep into it.

Maybe better to hold off

for the dot one.

But I just wonder about

like if encapsulation,

so if it's,

if something's fully

informationally encapsulated,

like, you know,

my heart is fully

informationally encapsulated

from my brain,

like,

how do you,

how does the living system

function, right?

So,

so this partial

encapsulation,

if that's not strong

enough for modularity,

like,

I don't know how full

encapsulation is possible.

I just,

I don't see it.

So.

Yes.

Section three

is entitled

A Golden Thread,
Re-Energy Principle.

And so,

we just had to go there
with a Blake reference
because on his gravestone
and one of his most famous
lines is,

I give you the end
of a golden string,
only wind it into a ball.

It will lead you
in at Heaven's Gate,
built in Jerusalem's wall.

And then here's a drawing
that he made
with a mythological character
walking with a golden string.

So the general idea,
according to Benny,
behind predictive coding
is that the brain
uses approximate
Bayesian inference
to decrease discrepancy
between predictions
and inputs.

That is one of the
key contentions
of the FEP.

And the organism's
interaction with its
environment can be
represented by Markov blankets.

Here, I am using
representation and modeling
when speaking of Markov blankets

and consider them
to be Markovian models.

So, check out

Benny's

guest stream appearance.

I think it's number one
on Markovian monism.

And

this sentence makes it clear
we're talking about
models.

We're in the realm
of instrumentalism.

We're talking about

Markovian models,

not about whether

we're making an

ontologically real claim

about Markov blankets

like being the structure
of reality.

So in these models,

Markov blankets

are describing

the conditional independence

of internal

and external states.

And then it's that

Friston innovation

beyond Perl

and Markov's

to describe

the incoming

dependencies as sensory

and the outgoing

dependencies as active states.

So,

what is the golden thread?

brain

I can't find it
now in here.

Anyway,

I think that
there's a subtle
and a nuanced
difference between
what FEP

is as a process
and as a principle
and what
FEP

architecture
is.

And I think
that is
what he is
trying to bring
to the table.

There is a
difference between
the architecture
or the product
that results
from the use
of an FEP
process
and the
FEP
process itself.

And once I
figure out
my own notes
here,
I can speak
to what that

BCC

is.

But I'll

shut up

for a second.

Very well.

Interesting

to describe

the difference

between the

principle

and the

realized

architecture.

Section four.

The track

of a storm.

So here's

another

single

slider.

So there's

some apparent

tension between

the modularity

thesis and the

predictive

architecture.

So that's

what

Kirchhoff and

Hippolito

described.

While the

thesis of

modularity

points in

the direction

of segregation
and encapsulation,
the predictive
architecture grounds
the integration
of information
in a hierarchical
top-down
architecture that
unifies the
mechanisms of
cognition,
perception,
and action
under the
FEP.

So it's like
what Blue
just said.

Wait,
if it's
encapsulated
and defined
by its
separation,
then how
is it part
of this
integrated
whole?

Well,
Howey in
2013,
the book
The Predictive
Mind,
and other
people in

other places
have suggested
that this
conditional
independence
between the
various levels
of the
hierarchical
structure of
predictive
coding under
FEP could
be construed
along the
lines of
functional
segregation
and
informational
encapsulation,
which is why
Fodor was
brought up
earlier.

Dresen
argued that
the predictive
architecture can
be modeled in
terms of this
causal probabilistic
dependency where
different nodes
could influence
the adjacent
levels,
but there isn't

global transitivity.

So like the
lowermost nodes
are not
influencing all,
just the ones
directly above
them.

However,
the Hippolito
and Kirchhoff
paper challenge
that vertical
and horizontal
account.

So the vertical
account is like
the multi-scale
nested systems,
the horizontal
account is like
the collective
behavior interactions
across within a
level, and so
they aim to
refute this
account of
functional
informational
modularity.

According to
Hippolito and
Kirchhoff, and
let's hear from
them if it's not
the case, the
intransitivity

argument for
modularity is
based on DAGs.
However, they're
going to argue
that DAGs are
not suitable
for that kind
of modeling
because the
brain has
these cyclic
mechanisms, and
so DAGs have
to be replaced
by cyclic
models, which
don't have the
limitations of
DAGs.

And then with
respect to the
Markov blanket
point, Benny
asserts that
Hippolito and
Kirchhoff argue
that the argument
from Markov
blankets is of
little avail to
establishing
modularity.
He will get
to both of
those arguments.
He kind of
doesn't really

worry about the
courtroom as
much, but the
intransitivity and
the Markovian
blanket arguments
are what Benny's
going to focus
on.

But first,
remark on the
importance of
scientific models.

So, kind of a
fun tack, but
that's section
four.

Yes, please,
Dean.

Yeah, so I
found out BCC
is brain's
cognitive
capacities.

That's how he
described it in
the paper.

So, I don't
think it's a
rock, paper,
scissors issue
either.

I believe that
predictive coding
is entailed by
the brain's
cognitive
capacities.

I believe that
the brain's
cognitive
capacities are
entailed by
the free energy
principle.

So, it
sounds like
you've got a
rock, paper,
scissors, but
that's not the
same as the
free energy
principles
architecture
being entailed
by predictive
coding.

I would have to
diagram that out
for you, but
that's where he's
saying it's not
an intransitivity
argument.

And that's just
notes to me,
and that's where

I kind of ran
into his
corner, even
though I wanted
to actually do a
paper later this
year, a review
of one of

Hippolyta's
papers on an
action that I
think is
fantastic in
instrumentalism.

But this was
the moment where
it was clear to
me that it's
not an
intransitivity
argument.

Because back to
what Axel
pointed out to,
we have to
know what
entails what.

Very cool.

Section five.

A knock at
the door.

Models and
modeling.

We really do
need to do
that joke
stream.

Knock, knock
jokes.

Like, you
know, who's
there?

I don't know.

Who are you
expecting?

Who do you

prefer?

In this

section,

Benny is going

to describe

some comments

on modeling.

He writes,

I do not

think we can

find a

reliable model

independent

handle on the

architecture of

the brain.

So it's kind of

like saying, if

we're going to

science, we're

going to

model.

So let's not

think that we

could science

without modeling.

The general

insight is that

the interface of

scientific theories

and their target

systems takes

place through

scientific modeling.

So theories

about the

brain, the

brain itself, and

then the edge
is the scientific
modeling.
Now, the brain
is like a
special and
fascinating system
because it can
be reflexively
modeling itself.
But if you
think about
theories of
pendulism,
pendulums,
theory of
pendulum, and
then the
target system
of the actual
pendulum, it's
pretty clear to
see where the
modeling relation
comes into play.
However, again,
it gets kind
of strange
loopy with
the brain.
And Benny is
drawing upon
various works.
Some of it is
the Weisberg
work on
like simulation
and similarity

using models to understand the world.

And also some of Benny's earlier work, for example, the ways in which Markov blankets are construed as scientific models.

And similar points have been discussed by Mel Andrews 2021.

That was Actim 14.

So scientists are making and applying models, and so we're not going to break out of that VIP room in Plato's cave.

So we have different kinds of models, and maybe some of them are preferable, but like if your argument is predicated on

breaking free of
a given
scientific model,
it is
something to
hear if you
think you have
it.

Dean?

So this is,
it got
interesting for
me here too,
because if you
look, for
example, at a
slice of time
in an
fMRI image,
you could be
focused on
the concentration
of heat in
different areas
of the brain.

That would be
one bit of
information that
you could take
away.

or you
could look
at the
proportion of
blood in
different areas
of the brain,
which is going

to tell you a
little bit of
different information
than the actual
concentration of
heat, that
heat bump per
se.

And so I was
kind of curious
who is a
modeler here,
because he
talks about we
can't take
the observer
out of the
observation.

So again,
what relative
bit of
information do
we tend to
hone in on,
or do we
think is the
more priority
information to
take away from
one of these
slices of time
representations?

That was really
important to me
right here,
because if you
think of
encapsulation as

that sort of
frenetic image a
few slides back
where all the
brain is chopped
up like,
here's the
bacon and
here's the
ham and
the different
parts of the
pig,
you're choosing
what it is that
you think is
the way of
encapsulating,
but I don't
think that's
what he's
talking about.

We'll
explore it,
but yes.
Taking this
kind of a
really strong
principled stance
on scientific
modeling means
that we can
always ask,
who is doing
the modeling?
Why?
And so actually
it's not just

this sterile
philosophical
point, it
really brings
us to consider
the social and
the operational
aspects of
science.

So it's a
very important
sociotechnical
point.

Section 6,
the game
made, the
intransitivity
argument.

Okay, we're
not going to
go through
all the
details here,
we might
have time
in the
dot 1 and
2, but
there, Benny
is going to
focus on
Hippolito and
Kirchhoff's
specific reply
to the
intransitivity
argument that
is centering

the distinction
between the
directed
acyclic
graphs and
directed
cyclic
graphs.

Benny is
going to
argue that
these two
kinds of
statistical
models do
different
things.

It turns
out they
actually also
map on to
the difference
between functional
and effective
connectivity.

And so
applying one
sort of the
model DAG or
the cyclic
model in a
specific situation
is a matter of
practical exigency
and methodological
consideration as well
as the goals and the
interest of the

modeler.

So it's like

we're 100

miles down

choosing a

model freeway

and then

whether you

take this

slight difference

between one

topology of

graphical model

that's quantitative

versus a

different topology,

maybe that is

not going to

be super

philosophically

impactful because

we already have

taken like the

big pill,

which was

making a

scientific model

or making a

graphical

quantitative

model.

There's a

citation to

the Fristin

Visa

Hobson

2020 paper

that also

describes like
where the
cyclic versus
acyclic are
preferable.

And

Benny

recaps and

just basically

to summarize,

there is no

universal reason

for extolling

DAGs over

DCMs because

the focus on

functional

connectivity or

effective

connectivity

is a matter

of explanatory

predictive goals

and interests.

So,

scientists are

using different

kinds of

models.

And so,

maybe we

should pull

back to

scientists

use models

and think

about the

implications of

that rather
than try to
finesse out
some philosophical
implications from
the very
specific kind
of last mile
modeling choices.

DCMs can
do certain
things,
whereas DAGs
can do
other things.

Which one of
these approaches
can be used to
analyze data
in the context
of the FEP?

We're going to
discuss it
in 35.1.

Benny says,
I do not
think
Hippolito and
Kirchhoff 2019
are bound to
concede a
non-realist
reading of
DCM.

Rather,
the critique
of how
H&K

confute the
intransitivity
argument is
pointed at
their unjustified
partiality towards
DCMs.

So,
it's like,
if you could
just scapegoat
the DAG,
then the
DCM is the
savior.

But,
if both of
them are just
types of
scientific models
that are chosen
situationally,
operationally,
as Fristin
described,
then,
there is no
savior model
coming in to
save the day.

DCMs are
formal devices.

And then,
there's a citation
to the
emperor's new
Markov blankets
with Yella et

all,
livestream number
20.
And so,
drawing attention
to the difference
between DAGs
and DCMs,
which are
complementary
modeling tools,
as we discussed
earlier,
they're defined
as complementary,
could not
confute the
intransitivity
argument.
So,
if we have
already committed
to the
instrumental
perspective,
so we're
talking about
our model,
not about
the world,
then,
the difference
between
different modeling
approaches is
purely
situational or
operational.

And so,
all scientific
models have
roughly similar
philosophical
grounding,
or they're
part of a
broad category,
with respect
to this
intransitivity
argument.

Nice
philosophy
and
interesting
points.

Okay,
column in the
storm.

The
indispensability
of modularity
and the
indispensability
of scientific
models.

Oh,
I wonder
if there's
going to
be like
a parallelism
between
models and
modularity.

And here's

like a

person in

their

Markov

bubble

in the

storm.

This section

focuses on

a realist

interpretation

of DCM

in terms of

dynamical

causal models.

So,

even

instrumentalists

can

but both

ways.

You know,

we're not

obliged to

never talk

about stances

that in

one certain

situation we

didn't

prefer.

It's just

really

wide-ranging

intellectual

exploration

that's being

characterized

here.

And here,

Benny does

play on a

little bit of

an acronym

play,

which is that

DCM

applies to

dynamic causal

modeling as

well as

directed

cyclic

modeling.

So,

cyclic

graphs or

the dynamic

causal

graph.

And he'll

argue through

an example

that we're

going to

discuss not

today with

Alice and

Bob.

He argues

that the

modularity in

the context

of DCM,

it's not as

strong as

the Fodorian
isolationist
concept,
but in
interesting and
important ways,
it's going to
be enough.

And so,
a moderate
form of
modularity is
retained.

And
Hippolyto and
Kirchhoff argue
that this
weakened notion
of modularity
is explanatorily
vacuous,
it's empty,
and there would
be no reason
for keeping it.

And that's
the bone of
contention.

He's going to
dissent and
say that the
benefit of this
modest notion of
modularity is
remarkable.

So,
that's the
claim.

Any comments
on section
seven?

What would
happen if we
couldn't
categorize?

Then,
it would be
impossible to
give a
categorical
response to
that question.

And a lot
more, right?

Like, that's
basically what
this comes down
to.

It doesn't
mean that
things aren't
arbitrary and
not necessarily
put into the
correct silos,
but there is a
great benefit.

I don't think
it's dispensable
our capacity to
categorize.

It doesn't mean
we always are
obliged to do
it, but when
we do do it,

is there a
benefit?

Yes.

Is there a
cost?

Maybe.

But if we
took it away,
what would
the world
look like
then?

a triangle

is a
triangle

is a
triangle,
right?

There would
be no
specificity
then.

Interesting.

We'll talk
more about
categorization
models and
the parable
of Alice
and Bob.

Section 8,
still knitting,
Markov
blankets.

So here
they return
to this
idea of

the vertical
insulation
and the
horizontal
insulation.

Like there's
encapsulation
laterally,
collective
behavior,
and then
encapsulation
like an
onion,
like a
multi-scale
system.

So that's
very interesting.
and this
is also
where we
get to
this point
about the
scale-free
or scale-friendly
nature of
Markov
blankets.

The main
reason that
H and K
resist this
argument about
the Markov
blankets is
based on the

fact that the
regimentation of
states into
Markov blankets
is scale-free.

So it can
range across
scales.

Maybe even
that is
already scale-friendly.

Because Markov
blankets are
scale-free,
they do not
imply insulation,
and thus the
modularity thesis
does not receive
support from
them.

And so
Benny concedes,
yeah, it's
true.

Markov blankets
can be applied
to basically
a wide range
of phenomena.

However,
this is
overlooking a
subtle point.

Markov
blankets in
the abstraction
in principle

are scale-free.

There's no
inherent constraint
on their size
and scope.

Linear models
are inherently
a priori
scale-free.

It could be
centimeters,
versus the
GDP.

It could be
astronomical units
versus temperature.

Like, it
doesn't have a
tie-in to a
certain spatial-temporal
scale or any
feature for a
linear model or
for a Markov
blanket.

However,
cue Dean
from stage
left or, you
know, running in
from the audience.

Their application
or imputation
to various target
systems will impose
a definite scale
on Markov
blankets.

To make a long story short, in the context of their application under FEP, not in the context of FEP, Markov blankets are not necessarily scale-free mathematical models because they're being used in models.

So, even though the formal tools could be scale-free, application mandates calibrating to a certain scale.

If that's the case, and it seems like it is, then counter-arguments based upon the scale-free nature of Markov blankets could not undermine

arguments from
Markovian
formalisms
soundly.

So, this is
really nicely
framed and
placed in
space and
time.

Dean, what
do you think
about that?

Well, I'm
still, as I
said, I
didn't, first
of all, I
want to make
sure to
disclaim this.

I never
saw this
paper until
three days
ago, four
days ago.

So, my
guess, my
WAG about
scale-friendly
and sort
of whether
that was
something we
should consider,
it was nice
to read this

and have
somebody else
go, maybe
it's something
we should
consider.
I'm going
to leave
it at that
because I
don't know
if somebody's
going to
come along
and disprove
that and,
you know,
because if
that's,
if it's
falsifiable
that
calibrating
doesn't
matter and
categorizing
doesn't
matter, I'm
open to
hearing that
too.
But in
the meantime,
I think it's
something that
if we don't
recognize the
difference between

the abstraction
and that
thing that
we're focusing
on right now,
we can get
lost in the
idea that a
triangle is a
triangle is a
triangle and
nothing else
matters.

blue, anything
you want to
add on this
section?

Yeah, I
mean, I
really think
that, I
mean, just
hits the nail
on the head.

Like, we're
the ones
like putting
the Markov
blanket on
something.

And so,
like, just
because something
can be, you
know, modeled
using a Markov
blanket doesn't
necessarily mean

it doesn't

scale.

And, like,

things don't

Markov blanket

themselves,

right?

Like, so, I

mean, even

like my layer

of skin, like

cells are popping

off and new

cells are adding

on all the

time.

And so, like,

there's, it's

not a hard

boundary.

It's not a

solid boundary.

It's very

fluid, like,

what goes in

and out of a

Markov blanket.

So, I think,

like, our

imputation of a

Markov blanket

on a system, I

mean, yeah,

they're scale-free

means you can

impute them on

any system at

any scale.

So, so,
maybe scale
friendly is,
is the way
to go, but
it's not like
the blanket
itself is scale-free.

I mean, all
blankets are,
are composed of
stitches, right?

Like, so, are
you using big
stitches or
little stitches
or like
different layers
of cloth or
what?

So, I don't
know.

It's, it's,
I don't know.

I think it
really drives
it home.

Yeah.

And the paper
was, I mean,
we can double
check, but it's
from December
21.

So, we have
proof of
live stream,
Dean, that,

you know,
you were there.

Great.

Can I just
say one last,
because I
completely agree
with what
Blue just
said.

I was, I
was never
of an opinion
that, that
Markov blanket
isn't scale-free.

Of course
it is.

And it's
kind of like
saying there
are subject
matter experts.

Of course
there are,
but as we
had in the,
in the
conversation
with Dr.

Friston,
you could
also say
that there's
a prediction
matter expert.

Like, all
I'm asking

is for people
to consider
that there
could be
both, and
they could
be running
in parallel.

That's, I
wasn't trying
to push back
in the sense
that, you
know, people
are wrong.

What I was
saying is,
what, what
else could
we include
to fill,
fill in
the wholeness
of what
is happening
here.

That's all
I was trying
to point
out.

And again,
we could
take active
inference and
focus it
completely on
the minimizing
of free energy

and getting
what we're
attending to,
or we can
talk about
the availability
piece and
what, for
example, when
we're having
new people
on board
to our lab,
how, how
we make
that, that
entry point
so, so
relationship
based.

We could
do that.

We could
make, for
them, we
could make
them prediction
matter experts
before they
even realize
it in
themselves.

Now, whether
we do that
or not, it's
not up to
me, but we
could.

So that's
what I was
trying to
bring up, and
that's what I
like about
this paper.
great.

Last section,
last content
slide.

Isn't it
something they
say, like, if
the newspaper
article ends
with a
question mark,
I forget
which one.

Is it the
it's always
yes or it's
always no,
but the
footsteps die
out forever.

So this is a
summary in the
ninth section.

Benny
challenges the
main points of
Hippolito and
Kirchhoff.

Hippolito and
Kirchhoff
confuted the

modularity thesis.

So Benny is salvaging the modularity thesis from the pincer attack of Hippolito and Kirchhoff.

To support the argumentation of the paper, Benny draws upon the indispensability of a model-based conception of modularity.

That's where that kind of science and society angle and philosophy angle came into play.

Benny argues that DAGs and DCMs are complementary models and the choice between them is operational.

It's pragmatic.

It's not metaphysical per se.

So modeling is situational.

We can take a

deflationary
approach even
to how we
think about
science.

Benny argues
that when we
think about
dynamic causal
modeling,
there is
modularity
because of
how the
statistical model
pulls out
differential
coupling between
brain areas.

And that's
not a statement
about the
brain.

It's a
statement about
the model.

Benny also
challenges
Hippolito and
Kirchhoff's
answer to the
argument from
the Markovian
formalism by
showing that
although these
models are in
principle scale-free

in order to
represent any
specific target
system, they
have to be
associated with
interpretations
that do give
definite scales.

So the
Markov blanket
formalism is
conceptually
scale-free, but
any realization
application is
scale-specific,
thus scale-friendly,
tickles 21 at
best.

Personal
communication.

And basically...

I'll be
citing the
live stream
soon.

Yeah.

Benny closes with
just saying it
doesn't undermine
the claims and
this isn't like a
radical pro-modular
thesis.

It's not the
isolation of
brain regions and

they're each like
virtual machines that
are totally
separated.

The only point
that comes to
light by our
enterprise in this
paper is that the
debate over
modularity needs
to take the role
of scientific
models into
account and a
model relative
conception of
modularity cannot
be waved away
easily.

So it is a
great service and
an interesting
paper in form
and function.

we have an
empty slide for
the implications
and questions.

We have the
early slide
with what
we're going to
discuss in
35.1 and
35.2.

And it
was an awesome

discussion.

So if you

have any final
comments, now

would be a

good time.

do you think

that when you

get people who

have worked

together as

this author

and the two

people that he

was kind of

pushing back

against, do you

think when they

take it into

this kind of a

formal setting of

paper writing

that that changes

their relationship

fundamentally?

that's one of

I mean, I

suppose we could

only ask the

authors themselves,

but does it

have to be a

debate necessarily?

Does it have to

be drama

necessarily?

Or is there

kind of a

moving forward
hand in hand
part to this
I'm not
seeing?
It's a really
interesting question.
I'm sure
scientists and
all kinds of
different people
are going to
have different
answers, but
I think about
the papers in
their exact
phrasing as
like pheromones,
digital stigma
G in the
literature corpus.
And then the
person is like
the nest mate
who is weaving
on.
So any artifact
is going to be
made by that
person in the
past.
So I could be
like talking to
blue and it's
like, oh yeah,
you know,
night 2018 said

this or, you
know, but you
just said this,
but, but didn't
night 2009 say
this?

It's not a
contradiction.

It's just like
the trajectory and
the pheromone
deposition of
active and
complex entities.

And so I think
that that I'd
hope is the best
of both worlds,
which is we can
humanize Kirchhoff
and Hippolito
and Benny,
all the humans
involved and
take like a
compassionate and
even friendly
tack while also
being very
scrutinizing of
the specifics of
the knowledge
artifacts that
people leave.

Because a lot of
times there is
like a gray zone
where it'll say

something in text,
but then, oh,
if you're in the
club, then you
know that they
actually meant
something else or
you got to read
between the lines
with the methods
or you have to
read between the
lines with the
citations.

And that's kind
of a conflation
of the person
and the arguments.

Like when people
give an overly
charitable reading
or even try to
insinuate things
that weren't there
in the initial
text because they
want to uphold
their legend
of person.

and rather we
can just kind
of separate
those two in
how we model
the generative
process and
the generated
product.

I just think
that there's so
much you can
learn when
you're wrong.
And that doesn't
mean you want
to be a serial
mistake maker,
but I don't
think anybody
that's publishing
is looking to
do that either.
But if you're
comfortable, if
you're actually
comfortable with
putting wrong
answers out there
and having put
people kind of
prove that you're
wrong, but you're
comfortable with
that, I think
that makes a
huge difference,
at least in terms
of the culture
that you're
participating in.
It's a lot
safer if you
can throw
stuff out.
In four weeks
or five weeks

time, somebody
can say, no,
Dean, you were
wrong.

We've got to
take that
citation down.

I wouldn't
take it down.

I would leave
it up.

I would simply
say, no, in
that moment, I
thought I was
right, and now
I've updated
my generative
model.

Yeah, it's
like showing
your work.

Here's what
I knew at
this time, and
this is what
leads me to
my conclusions.

And then
someone says,
well, you
missed this
paper from
1855.

Okay.

Then that
updates the
next round of

pheromone

deposition.

Yeah, exactly.

Blue, any

last comments?

Good.

Yeah, it's a

fun start to

22.

And it's the

two architectures,

2022, second

year of ActInf

Lab, 22222222 is

coming up.

I mean, it

couldn't get any

better.

That's great.

Thank you,

folks.

Yeah, so see you

and everyone

in the